

SOS + DocMind

Complete Fix Plan — Phase 1 & 2

Everything needed to get the app running and stable

This document contains the exact code for every change. Follow fixes in order. Each fix is self-contained — the file, the problem, the old code, the new code, and why.

Fix Summary

#	What	File	Phase
1	deleteById → deleteByLectureId (table corruption)	AIOutputRepository + AIOutputService	P1 — Critical
2	Wrong @Param import (all typed AI GETs → 500)	AIOutputRepository	P1 — Critical
3	Raw entity serialization → LazyInitializationException	AIOutputController + AIOutputService	P1 — Critical
4	Frontend AIOutput interface field mismatches	saiOutputService.ts	P1 — Critical
5	DocMind save_index indentation bug	api/main.py	P1 — Critical
6	processLecture — no transaction boundary	LectureService	P1 — Critical
7	RevisionService — N+1 lazy load ownership check	RevisionService	P2 — Stability
8	Revision anchor date wrong (upload vs process)	RevisionService	P2 — Stability
9	DashboardService — undefined cache name	CacheConfig + application.properties	P2 — Stability
10	Flyway + ddl-auto=update conflict	pom.xml + application.properties	P2 — Stability
11	assertOwnership throws 404 instead of 403	LectureService + RevisionService	P2 — Stability
12	label_generator abstain (-1) skews importance	label_generator.py	P2 — Correctness
13	sentence_splitter returns wrong list	sentence_splitter.py	P2 — Correctness
14	summarizer joins sentences in score order, not order	summarizer.py	P2 — Correctness

Phase 1 — Critical Fixes (Do These First)

The app cannot function correctly without all six of these. Fix them in order — Fix 3 depends on Fix 2, and Fix 4 depends on Fix 3.

Fix 1 — deleteById deletes the wrong row — table corruption on every reprocess [CRITICAL]

File: AIOutputRepository.java + AIOutputService.java

Problem: saveAIOutputs calls aiOutputRepository.deleteById(lecture.getId()). deleteById takes an AI_Output primary key, not a lecture ID. It silently deletes a random row from a different lecture. Every reprocess duplicates rows; query results become nondeterministic.

Step 1 — Add deleteByLectureId to AIOutputRepository.java

Remove the wrong Lettuce @Param import. Add the correct Spring Data import and a derived delete method. The full file should be:

```
package com.Ajwain.SOS.repositories;

import java.util.List;
import org.springframework.data.jpa.repository.JpaRepository;
import org.springframework.data.jpa.repository.Query;
import org.springframework.data.repository.query.Param;
import org.springframework.stereotype.Repository;
import com.Ajwain.SOS.entities.AI_Output;
import com.Ajwain.SOS.entities.enums.OutputType;

@Repository
public interface AIOutputRepository extends JpaRepository {

    List findByLectureId(long lectureId);

    void deleteByLectureId(long lectureId);

    @Query("SELECT a FROM AI_Output a WHERE ... ")
    AI_Output findByLectureIdAndAiOutputType(
        @Param("lectureId") Long lectureId,
        @Param("type") OutputType type
    );
}
```

Step 2 — Fix saveAIOutputs in AIOutputService.java

Replace deleteById with deleteByLectureId:

```
public void saveAIOutputs(Lecture lecture, AIResponseDTO response) {
    aiOutputRepository.deleteByLectureId(lecture.getId());
    saveAIOutput(lecture, response);
}
```

Also delete the entire commented-out block (the old `generateAIOutputsForLecture` implementation) between the constructor and the new method.

Fix 2 — Wrong `@Param` import [CRITICAL]

File: `AIOutputRepository.java`

This fix is already done inside Fix 1 above. The full `AIOutputRepository.java` shown in Fix 1 is the correct final state.

Fix 3 — Raw entity → `LazyInitializationException` [CRITICAL]

File: `AIOutputController.java` + `AIOutputService.java`

Solution: Introduce `AIOutputDTO` and map inside the service. See full document for complete code implementations.

Fix 4 — Frontend field name mismatch [CRITICAL]

File: `src/services/aiOutputService.ts`

Update interface to use `outputType`, `outputContent`, `generatedAt` to match backend `AIOutputDTO`. See full document.

Fix 5 — DocMind indentation bug [CRITICAL]

File: `DocMind-Experimental/api/main.py`

The closing `}` of `indexes dict` and `save_index` call must be inside the `try` block with proper indentation. See full document.

Fix 6 — `processLecture` without `@Transactional` [CRITICAL]

File: `LectureService.java`

Add `@Transactional` to `persistProcessingResults()` to ensure all DB writes commit together or rollback on failure. See full document.

Phase 2 — Stability Fixes

These fixes prevent crashes, silent failures, and incorrect data. Do them before real users hit the system.

Fix 7 — N+1 lazy loads in getRevisionsForLecture

File: RevisionService.java

Use `findByLectureIdAndLectureSubjectUserId` to fetch with ownership check in single query.

Fix 8 — Revision schedule anchored to upload date

File: RevisionService.java

Change `lecture.getUploadDate()` to `LocalDate.now()` in `createRevisionSchedule`.

Fix 9 — Redis not configured / undefined cache

File: application.properties + CacheConfig.java

Add Redis properties and register dashboard cache with 2-min TTL.

Fix 10 — Flyway conflict with ddl-auto=update

File: application.properties

Set `spring.flyway.enabled=false` until migration scripts are ready.

Fix 11 — assertOwnership throws 404 instead of 403

File: LectureService + RevisionService

Create `UnauthorizedException` and handle with 403 status.

Fix 12 — label_generator abstain values skew scores

File: label_generator.py

Mask abstain votes (-1) before averaging with `np.ma.masked_equal`.

Fix 13 — sentence_splitter returns unfiltered list

File: sentence_splitter.py

Return `fsentences` (filtered) not `sentences`. Fix regex: `! not |`.

Fix 14 — summarizer joins in wrong order

File: summarizer.py

Sort selected sentences by original document position before joining.

Environment Setup Checklist

Before starting the app, make sure these environment variables are set:

Variable	Example Value	Notes
DB_PASSWORD	yourMySQLPassword	Required. No default.
JWT_SECRET	a-string-at-least-32-chars-long!!	Required. ≥32 chars for HS256.
DOCMIND_URL	http://127.0.0.1:8000/api	Optional. Defaults to localhost:8000.
SOS_STORAGE_PATH	./sos_files	Optional. Where PDFs are stored.
REDIS_HOST	localhost	Optional. Defaults to localhost.
REDIS_PORT	6379	Optional. Defaults to 6379.

Start order: (1) MySQL, (2) Redis, (3) DocMind uvicorn, (4) Spring Boot, (5) React frontend.

Startup Commands

DocMind:

```
pip install -r requirements.txt
uvicorn api.main:app --host 0.0.0.0 --port 8000 --reload
```

Spring Boot:

```
export DB_PASSWORD=yourPassword
export JWT_SECRET=your-32-char-minimum-secret!!
./mvnw spring-boot:run
```

Frontend:

```
npm install && npm run dev
```

Verification Checks

1. GET http://localhost:8000/ — should return DocMind status
2. GET http://localhost:8080/api/auth/login — should return JWT
3. POST /api/lectures/{subjectId} with PDF — should return 201
4. POST /api/lectures/{lectureId}/process — should return 200
5. GET /api/ai-output/lecture/{lectureId}/summary — should return AIOutputDTO
6. POST /api/ai-output/query — should return results